PIPELINE AUDIT AND HEALTH REPORT

Mixed products: routing rules, priorities, changeovers

Orders of types A/B/C are sorted by rule: A to line A, B and C to line B (which serves high-priority C first). Both lines feed a shared paint booth that pays a changeover whenever the part type changes. Try raising the paint booth's speed or lowering its setup time.

Generated 2026-10-13 19:48 | simulated 2.0 h | seed 11 | 10 replications

HEALTH SCORE

94 /100

Grade A

Healthy: the design is robust and has headroom.

Critical 0 | High 0 | Medium 1 | Low 1 | Info 2

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
164.5 units/h	329 units	4.0%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
6.9 units	149 s	line_a

Executive summary

Primary bottleneck: 'line_a' (73% busy, 10% down, index 87). 4.0% of arrivals are lost. Run what-if to measure which lever recovers the most throughput.

What to fix first

- Medium (arrivals): Check whether the loss is acceptable for this process. Measured best lever: 10% faster processing at 'line_a' (+1.6% throughput).
- Low (throughput): Use a longer run if you need tighter estimates.

Best measured improvement

Best lever: 10% faster processing at 'line_a' -> +2.6 units/hour (1.6%), measured over 3 seeds with common random numbers.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
Medium	Flow	**arrivals**	4.0% of arrivals are lost. Action: Check whether the loss is acceptable for this process. Measured best lever: 10% faster processing at 'line_a' (+1.6% throughput).	-5

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
Low	Confidence	**throughput**	Throughput 95% CI is +-5.5% of the mean. Action: Use a longer run if you need tighter estimates.	-1
Info	Capacity	**line_a**	Constraint of the line: 83% utilised. Action: Improvements elsewhere will not raise throughput until this station is relieved. Measured best lever: 10% faster processing at 'line_a' (+1.6% throughput).	
Info	Reliability	**dependencies**	2 stations lie on every path (serial structure), but none has a failure model. Action: Fine for an ideal analysis; add failure data to find out how fragile this is.	

Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
orders	source	-	-	-	-	-
sorter	machine		9.3	14.3	0.0	0.8 / 7
line_a bottleneck	machine		73.2	1.7	9.9	2.4 / 5
line_b	machine		64.2	1.8	0.0	1.3 / 5
paint	machine		75.5	0.0	0.0	2.4 / 7
ship	sink	-	-	-	-	-

Work in process over time

Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	165.75 units/h	+- 9.16	12.80	5.5%
Average WIP	5.88 units	+- 0.88	1.24	15.0%
Cycle time	126.58 s	+- 17.63	24.64	13.9%
Arrivals lost	3.31 %	+- 2.53	3.54	76.6%

No replication deadlocked.

Failure model

NODE	MTBF	MTTR	AVAILABILITY	OBSERVED DOWN	FAILURES
line_a	30 min	2.5 min	92.3%	9.9%	4

Single points of failure

All units must pass through: sorter, paint. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 349 = completed 329 + scrapped 0 + lost 14 + in system 6
Little's Law	PASS	L = 6.90, lambda x W = 6.79 (gap 1.5%)
Run length	PASS	window = 48 x cycle time (30x or more recommended)
Warm-up	WARN	no warm-up: start-up transients are included in averages
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 169.1 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
line_a	10% faster processing	171.7	**+2.6**	+1.6%
line_a	20% shorter repairs	171.0	**+2.0**	+1.2%
line_a	+2 buffer slots	170.6	**+1.6**	+0.9%
line_b	10% faster processing	170.1	**+1.1**	+0.6%
line_b	+2 buffer slots	170.0	**+0.9**	+0.6%
paint	10% faster processing	169.5	**+0.4**	+0.3%

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
paint	+2 buffer slots	169.4	**+0.3**	+0.2%

Best lever: 10% faster processing at 'line_a' -> +2.6 units/hour (1.6%), measured over 3 seeds with common random numbers.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
orders	source	interarrival=exponential(20), mix={"A":0.6,"B":0.3,"C":0.1}, priority={"C":2}
sorter	machine	capacity=6, process=2
line_a	machine	capacity=4, mtbf=1800, mttr=150, process_by_type={"A":{"dist":"normal","mean":25,"stddev":3}}
line_b	machine	capacity=4, discipline=priority, process_by_type={"B":{"dist":"normal","mean":30,"stddev":4},"C":{"dist":"normal","mean":45,"stddev":6}}, setup=12
paint	machine	capacity=6, process=exponential(10), setup=8
ship	sink	(defaults)

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
Medium	**arrivals**	4.0% of arrivals are lost.	-5
Low	**throughput**	Throughput 95% CI is +-5.5% of the mean.	-1
	Total deductions		**-6**
	Health score		**94**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.